Agri-EconoBot

Powered by Team INNOX

Low-cost smart robot for seed & fertilizer spreading and basic crop monitoring. Designed for small and medium-scale farmers to reduce labour, lower input waste, and boost yields.

TEAM INNOX

Problem Statement

Small farmers face labour shortages and inconsistent manual spreading that wastes inputs and reduces yields. Mechanical options are often expensive or not suited to small plots.

Objective

Develop an affordable, autonomous robot that precisely spreads seed/fertiliser and monitors crops — saving time and increasing productivity.

Target users

Small & medium-scale farmers, extension officers, and local agri-projects.

Design Concept & Framework

Compact, lightweight chassis for easy movement between rows. Modular design to allow different dispensing modules and sensors.

Chassis

Lightweight low-centre-of-gravity body for stability on uneven fields.

Navigation

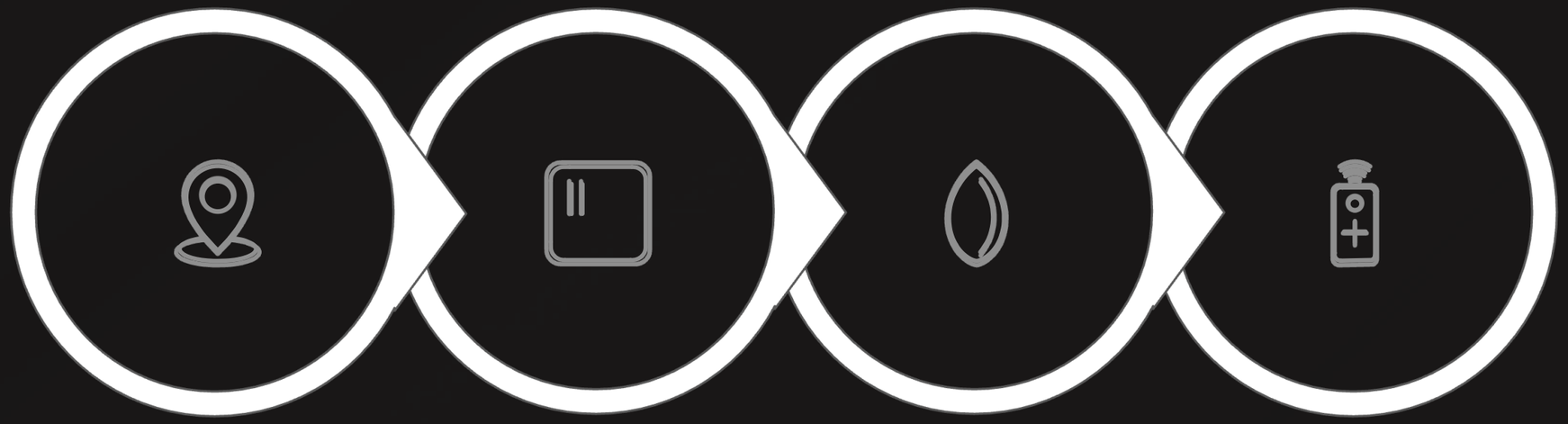
Row-guided movement using IR and ultrasonic sensors for obstacle detection and row alignment.

The Brain

Low-cost microcontroller(ESP32) manages motion, spreading schedules and basic telemetry to keep costs down.

Technical Specifications

- *Power: Solar-assisted charging + rechargeable battery for extended field use*
- *Sensors: Ultrasonic sensors for row following and obstacle avoidance*
- *Dispensing: Motorised hopper/tank with flow control for precise seed/fertiliser delivery*
- *Durability: Weather-resistant components for rain and dusty field conditions*



Position Align Dispense Override

The robot follows a simple, repeatable workflow. Sensors maintain alignment and speed while the hopper dispenses accurately; operators can intervene with a remote controller when needed.

Practicality & Cost Analysis

Designed for low-cost construction using locally available parts and straightforward assembly.

Build cost

Estimated between Rs. 7,000 and 10,000 depending on supplier and scale of purchase.

Market viability

Lower cost than repeated manual labour; attractive for cooperatives and smallholders.

Maintenance

Modular parts, simple motor replacements, and farmer-friendly controls minimise downtime.

Field Adaptability & Scalability

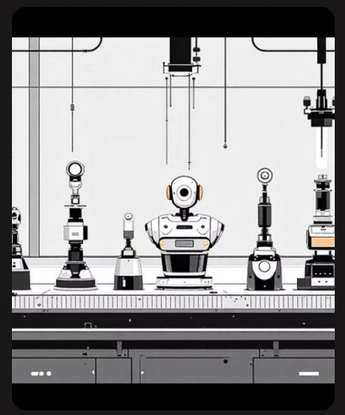
Optimised for crops with clear row structure (e.g., maize, vegetables, paddy). Settings are adjustable for row spacing, seed size and application rates.

Future module: Crystal Fertilizers

Add-on for adding crystal fertilizers and targeted pest control.

Future module: Weeding

Mechanical weeder attachment for inter-row weed control to reduce herbicide use.



Scalability

Mass production and local assembly can further reduce unit cost; cooperative ownership models recommended.

Expected Outcomes

- *Efficiency: Significant time and labour savings for routine tasks.*
- *Yield: More consistent application increases effective crop output.*
- *Energy & Waste: Solar assistance lowers power needs; precision reduces input waste.*

Conclusion & Next Steps

Prototype status

Prototype ready — functional demo available to show key principles and field performance.

Immediate next steps

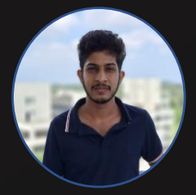
Field trials with target crops, refine dispensing calibration and sensor tuning, gather farmer feedback.

Longer-term goals

Finalise design for manufacturability, pursue patent protection and pilot adoption through extension programmes.

Meet the Team Behind the Robot

UDESHAANURANGA
Team Leader



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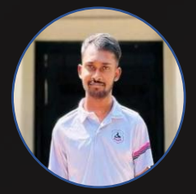
Mechanical Lead



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Thank you!